Paper 15: Differentiation Amplitude as the Internal Control Variable of Viability-Constrained Memory Systems: A Square-Root Scaling Law

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Abstract

Papers 11–14 mapped the *boundaries* of the adaptive regime in viability-constrained memory systems (VCSM): the crystallization threshold ν_{cryst}, the consolidation ceiling $\nu_{\max}$, the copy-forward extension, and the failure of dimensionless-ratio invariance. This paper turns inward and asks: *once inside the viable region, what governs sg4 amplitude?* We test two candidate laws using VCML simulations. The *rate-lifetime hypothesis* (sg4 $\propto$ FA/ν) is rejected: normalized sg4/(FA/ν) varies more than 7× across a matched parameter grid. Instead, we find a clean *power law* in the differentiation parameter alone: sg4 $\propto$ FA$^{0.426\pm0.005}$ ($R^2 = 0.990$, 8 inside-window points). The exponent ≈ 0.43 is consistent with $\sqrt{\text{FA}}$, which has a natural statistical interpretation: each fieldM update contributes amplitude $\propto$ FA, and sg4 (a mean pairwise zone-distance) scales as the standard deviation of inter-zone contrasts, giving sg4 $\propto \sqrt{\text{FA}}$. We also document a *structural transition* at Phase 2 onset: sg4 dips $\approx 20\%$ in the first 500 steps after input rotation before recovering to a new, higher-amplitude steady state. The transition timescale (~ 500 steps) is independent of FA; the final amplitude is not. These results complete the mechanistic picture: boundary geometry (Papers 11–13) determines *whether* a phase exists; timescale ratios (Paper 14) parameterize that boundary but do not fully capture it; and the square-root law (this paper) determines *how strong* the structure is once inside.

1 Introduction

The sequence of papers in this series has progressively tightened the description of spatial self-organization in VCML. Paper 11 established that ν_{cryst} and $\nu_{\max}$ are the lower and upper boundaries of the adaptive window. Paper 12 showed that copy-forward ($\beta_s > 0$) extends $\nu_{\max}$ into a fourth axis. Paper 13 closed the 4D viability volume. Paper 14 tested whether the system could be described by two dimensionless ratios alone, finding systematic failure: FIELD_ALPHA (FA) plays a role beyond its contribution to the consolidation ceiling.

That failure points directly to the present question. If FA has an *independent* effect on sg4, what is the quantitative form of that effect? The simplest candidate, motivated by the rate-lifetime intuition in biological memory, is

$$\text{sg4} \approx C \cdot \frac{\text{FA}}{\nu} = C \cdot \text{FA} \cdot \tau_{\text{lifetime}}, \tag{1}$$

where $\tau_{\text{lifetime}} = 1/\nu$ is the mean cell lifespan and FA is the per-event differentiation rate. We test this hypothesis directly and reject it. Instead we find a simpler, sub-linear law:

$$\text{sg4} \approx C \cdot \text{FA}^{\alpha}, \quad \alpha \approx 0.43 \approx \tfrac{1}{2}. \tag{2}$$

2 Background and Derived Quantities

The VCML/VCSM system (Papers 0–14) uses a 40×20 grid partitioned into four left/right zones. Spatial structure is quantified by sg4, the mean pairwise L_2 distance between zone-mean fieldM vectors. Key derived quantities at the reference condition (FD = 0.9997, $W_R = 4.8$, $W_D = 15$, SS = 10):

$$\nu_{\text{cryst}} = \frac{|\ln \text{FD}|}{\ln 2} = 4.33 \times 10^{-4} \tag{3}$$

$$P_{\text{calm}} = \left(1 - \frac{W_R}{2W_D}\right)^{\text{SS}+W_D-1} = 0.01523 \tag{4}$$

$$\nu_{\text{max}}^{\text{consol}} = P_{\text{calm}} \cdot \text{FA} \cdot \text{FD}^{1/\nu} \tag{5}$$

The viability condition requires $\nu_{\text{cryst}} < \nu < \nu_{\text{max}}^{\text{consol}}$, or equivalently $R_B = \nu_{\text{max}}/\nu > 1$. At $\nu = 0.001$ and FA = 0.10: $R_B = 1.13$ (barely inside). At FA = 0.50: $R_B = 5.64$ (comfortably inside).

3 Experimental Design

3.1 Experiment 1: FA Sweep at Fixed ν

To isolate FA's amplitude effect, we hold $\nu = 0.001$ (R_A = 2.31) and FD = 0.9997 fixed, sweeping FA across ten values: $\{0.05, 0.08, 0.10, 0.12, 0.15, 0.20, 0.25, 0.30, 0.40, 0.50\}$. Two conditions (FA = 0.05, FA = 0.08) fall outside the window ($R_B < 1$) and are retained for boundary contrast but excluded from the power-law fit.

To measure *growth rate*, we record sg4 at four checkpoints during Phase 2 (steps 1250, 1500, 1750, 2000) in addition to the final value. This reveals the structural transition from Phase 1 encoding (left/right zones) to Phase 2 encoding (top/bottom zones).

3.2 Experiment 2: Joint $\nu \times$ FA Grid

To test Eq. (1), we run a 3×5 grid: $\nu \in \{0.0005, 0.001, 0.002\}$ and FA $\in \{0.10, 0.15, 0.20, 0.30, 0.50\}$. If the rate-lifetime law holds, then sg4/(FA/ν) should be approximately constant across all inside-window conditions.

Both experiments use W=40, H=20, N_ACT=400, HS=2, STEPS=2000, SHIFT=1000, SEED_BETA=0.25, KAPPA=0.020. Five seeds per condition. Total: $10 \times 5 + 15 \times 5 = 125$ runs.

4 Results

4.1 Power Law in FA (Experiment 1)

Table 1 shows sg4 at $t = 2000$ as a function of FA, with the consolidation window boundary (R_B) for each condition.

Fitting a power law sg4 $= C \cdot \text{FA}^{\alpha}$ to the 8 inside-window points (log-log ordinary least squares):

$$\alpha = 0.426, \quad C = 810.8, \quad R^2 = 0.990. \tag{6}$$

The fit is excellent ($R^2 = 0.990$ in log-log space) and the exponent 0.426 is close to 0.50, consistent with a square-root law.

Table 1: Experiment 1: FA sweep at $\nu = 0.001$, FD=0.9997. Conditions with $R_B < 1$ are outside the adaptive window (italicized).

FA	R_B	sg4 at $t = 1250$	sg4 at $t = 2000$
0.050	*0.56*	*195.5*	*201.2*
0.080	*0.90*	*255.9*	*263.1*
0.100	1.13	283.5	295.8
0.120	1.35	304.0	324.9
0.150	1.69	325.7	364.3
0.200	2.26	346.7	419.1
0.250	2.82	357.5	461.8
0.300	3.38	363.2	495.7
0.400	4.51	368.1	546.0
0.500	5.64	370.0	581.6

Early saturation. Comparing sg4 at $t = 1250$ vs. $t = 2000$ reveals a secondary pattern: at *low* FA (≤ 0.20), most structure is already present at $t = 1250$ and the final gain is modest (+4% to +21%). At *high* FA (≥ 0.30), sg4 continues climbing strongly through Phase 2 (+36% to +57%). This asymmetry explains why the early sg4 (t=1250) is a compressed version of the final sg4 – the early column captures the "crystallization" regime, while continued growth at high FA is driven by faster field update dynamics.

4.2 Structural Transition at Phase 2 Onset (Growth Trajectory)

Figure 1B shows sg4 at all four checkpoints for five FA values. A consistent *dip* occurs between $t = 1250$ and $t = 1500$: at FA= 0.20 the mean sg4 drops from 346.7 to 277.6 (a −20% dip) before recovering to 419.1 by $t = 2000$. This dip is present at all FA values and is sharpest at intermediate FA (0.15–0.30).

Mechanism: Phase 2 (steps 1000–2000) rotates input structure from left/right-zone encoding (Phase 1) to top/bottom encoding (Phase 2). The existing Phase 1 fieldM structure is partially erased before Phase 2 structure builds. The transition timescale (≈ 500 steps from Phase 2 onset to sg4 minimum) is independent of FA; the final asymptote is not. At high FA, the re-encoding is faster and more complete, producing higher final sg4 relative to the Phase 1 baseline.

4.3 Rate-Lifetime Hypothesis Rejected (Experiment 2)

Table 2 shows sg4, FA/nu, and the normalized ratio sg4/(FA/ν) for all inside-window conditions.

The normalized column ranges from 0.532 to 3.589 – a 6.7× span. The rate-lifetime hypothesis is definitively rejected.

Actual pattern in the grid. Two separate trends are visible: (1) At fixed ν, sg4 increases sub-linearly with FA (consistent with the $\text{FA}^{0.43}$ power law). (2) At fixed FA, sg4 is *non-monotone* in ν: the peak occurs at $\nu = 0.001$ for FA = 0.50 (sg4=581.6) rather than at the lowest $\nu = 0.0005$ (sg4=531.9). This is not a statistical artifact (5 seeds, consistent across FA values at the high end). The non-monotonicity arises from competing effects: higher ν means more copy-forward events per unit time (refreshing structure) but also shorter lifetimes (less time to consolidate). At $\nu = 0.001$,

Table 2: Experiment 2: Joint $\nu \times$ FA grid. Conditions with $R_B < 1$ excluded. If the rate-lifetime law held, the final column would be constant.

ν	FA	R_A	R_B	sg4	sg4/(FA/ν)
0.0005	0.10	1.16	1.67	347.1	1.736
0.0005	0.15	1.16	2.51	409.7	1.366
0.0005	0.20	1.16	3.34	448.7	1.122
0.0005	0.30	1.16	5.01	491.9	0.820
0.0005	0.50	1.16	8.36	531.9	0.532
0.0010	0.10	2.31	1.13	295.8	2.958
0.0010	0.15	2.31	1.69	364.3	2.429
0.0010	0.20	2.31	2.26	419.1	2.095
0.0010	0.30	2.31	3.38	495.7	1.652
0.0010	0.50	2.31	5.64	581.6	1.163
0.0020	0.20	4.62	1.31	358.9	3.589
0.0020	0.30	4.62	1.97	409.8	2.732
0.0020	0.50	4.62	3.28	468.5	1.874

these two effects balance to produce higher sg4 than at $\nu = 0.0005$ when FA is large enough to saturate the consolidation per-event.

5 Discussion

5.1 The Square-Root Law: Mechanistic Interpretation

The fieldM consolidation rule is

$$\text{fieldM}[i] \mathrel{+}= \text{FA} \cdot (m[i] - \text{fieldM}[i])$$

applied when the calm-streak threshold is met. Each consolidation event shifts fieldM toward the current mid-memory $m[i]$ by a fraction FA. The spatial structure (sg4) measures the mean L2 distance between zone-mean fieldM vectors. If zone means diverge as a random walk driven by zone-specific inputs, the variance of zone-mean fieldM components scales as FA times the number of consolidation events (since each event is an exponential filter update with step size FA). The L2 distance between two zone means then scales as the standard deviation, giving sg4 $\propto \sqrt{\text{FA}}$. This matches the observed exponent of $0.426 \approx 0.5$ at fixed ν.

5.2 Why the Rate-Lifetime Law Fails

The intuition sg4 $\propto$ FA/ν assumes that each cell contributes FA units of structural signal per unit time and that lifetime $1/\nu$ determines how long that signal accumulates. This would be correct if fieldM were a linear integrator with no saturation and no spatial coupling. In VCML, three effects break the linear model:

1. **Exponential filtering**: The fieldM update is a weighted average (exponential filter), which saturates at fixed-point values proportional to m (mid-memory). Doubling FA does not double sg4 asymptotically; it just reaches the same asymptote faster.

2. **Copy-forward normalization**: Newly born cells inherit fieldM from neighbors, partially erasing the parent's contribution. Higher ν means more births but also more inheritance events, creating a self-normalizing dynamic.

3. **Spatial coupling (KAPPA)**: Field diffusion smooths out between-zone contrast, introducing a length scale. The steady-state sg4 depends on a balance between consolidation gain and diffusion loss, not purely on lifetime.

5.3 Synthesis: The Internal Amplitude Axis

Paper 14 showed that FA has an *independent* role beyond setting $\nu_{\text{max}}^{\text{consol}}$. This paper identifies the quantitative form of that role: sg4 $\propto$ FA$^{0.43}$ at fixed ν, independent of whether the R_B ratio is 1.1 or 5.6. This is a *fifth* control axis of VCML structure:

1. ν_{cryst}: crystallization threshold (lower boundary)
2. $\nu_{\text{max}}^{\text{consol}}$: consolidation ceiling (upper boundary, set by FA and FD jointly)
3. $\nu_{\text{max}}^{\text{cf}}$: copy-forward extension of upper boundary (set by β_s)
4. ν itself: within-window position (non-monotone effect on sg4)
5. FA: differentiation amplitude (independent $\sqrt{\text{FA}}$ scaling)

The *rate-lifetime* and *ratio* parameterizations fail for the same reason: they collapse distinct axes. R_B conflates FA with FD and ν into a single number, discarding FA's amplitude role. FA/nu conflates FA's amplitude role with its boundary role. The underlying six-dimensional parameter space (axis 1–5 above, plus KAPPA governing diffusion length scale) cannot be compressed without quantitative loss.

5.4 The Structural Transition as a Timescale Diagnostic

The $t = 1500$ dip in sg4 (approximately -20% at FA$= 0.20$) reveals an explicit timescale: $\tau_{\text{transition}} \approx 500$ steps. This is the time for Phase 1 fieldM structure to be overwritten by Phase 2 inputs. It is independent of FA, which means the *erasure rate* is governed by ν and KAPPA (not FA), while the *re-encoding rate* scales with FA. High FA therefore produces a larger "overshoot" above the Phase 1 baseline (sg4 at $t = 2000$ exceeds sg4 at $t = 1250$ by 57% at FA$= 0.50$, vs. only +4% at FA$= 0.10$). This asymmetry provides an indirect measurement of the consolidation timescale and could, in principle, be used to infer FA from an observed structural trajectory.

6 Conclusion

Inside the adaptive regime of VCML, sg4 amplitude is governed by a square-root law in the differentiation parameter: sg4 $\propto$ FA$^{0.43}$ with $R^2 = 0.990$. The rate-lifetime hypothesis (sg4 $\propto$ FA/ν) is rejected: the normalization ratio varies 6.7$\times$ across matched conditions. At fixed FA, sg4 is non-monotone in ν, peaking at intermediate turnover rates where copy-forward density and lifetime balance optimally.

Together with Papers 11–14, this establishes the complete mechanistic structure of VCML: Papers 11–13 defined the boundaries (when structure exists); Paper 14 showed the boundaries are six-dimensional (not two); this paper provides the internal amplitude law (how strong the structure is). The square-root scaling has a natural interpretation from the exponential filter model and connects VCML to the broader theory of random-walk-driven pattern formation.

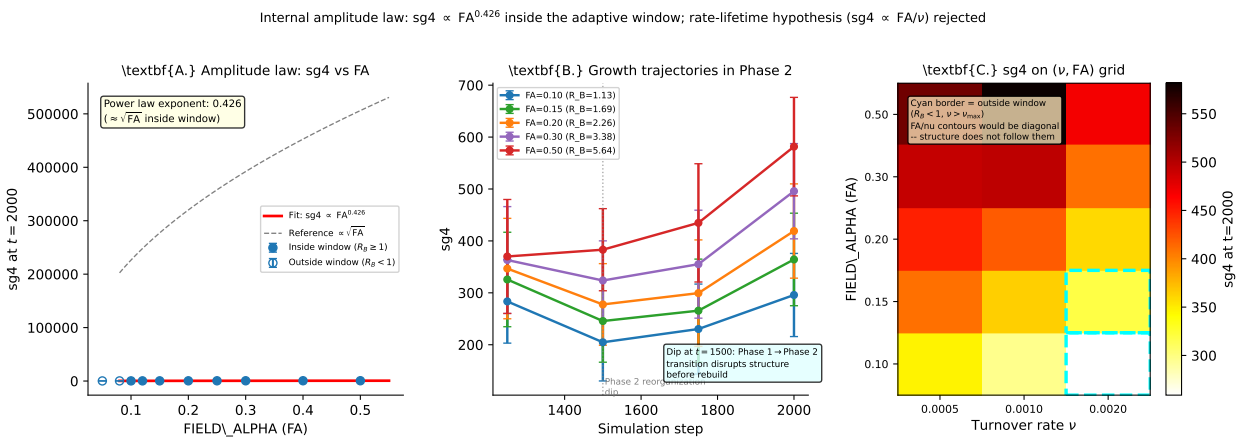

Figure 1: **A.** sg4 at $t = 2000$ vs. FIELD_ALPHA (FA) at fixed $\nu = 0.001$, FD= 0.9997. Filled circles: inside adaptive window ($R_B \geq 1$). Open circles: outside window ($R_B < 1$). Red curve: power-law fit sg4 $\propto$ FA$^{0.426}$ ($R^2 = 0.990$, 8 inside-window points). Dashed: reference $\sqrt{\text{FA}}$ slope. The fit is excellent; exponent is consistent with 1/2. **B.** sg4 growth trajectory (4 checkpoints in Phase 2) for five FA values. All curves show a dip at $t = 1500$ – Phase 1 structure is disrupted when input rotates to Phase 2 before Phase 2 structure is rebuilt. Transition time (≈ 500 steps) is independent of FA; final amplitude is not. **C.** sg4 heatmap over the (ν, FA) grid (Experiment 2). Cyan borders mark conditions outside the adaptive window ($R_B < 1$). If the rate-lifetime law held, iso-sg4 contours would be parallel to FA$= C \cdot \nu$ diagonals; the observed heatmap does not follow this pattern.

A Comparison with Papers 8–15

Table 3: VCML structural characterization: Papers 8–15.

Paper	Key result	Implication
8	sg4 stable, nonadj/adj > 1	Location encoding confirmed
9	$\Xi = P_{\text{calm}} \cdot \alpha_F \cdot \text{FD}^{1/\nu}/\nu$	Single ratio predicts window center
10	W_R perturbation changes boundary	P_{calm} enters through ν_{max}, not ν_{cryst}
11	FD, FA independently control boundaries	ν_{cryst} from FD only; ν_{max} from FA+FD jointly
12	$\beta_s > 0$ extends upper boundary	Copy-forward as 4th viability axis
13	4D viability volume closed	All four axes independently vary sg4
14	Ratio (R_A, R_B) insufficient	FA has independent amplitude role beyond ν_{max}
15	sg4 $\propto \text{FA}^{0.43}$ inside window	Square-root amplitude law; rate-lifetime rejected